Chapter 7 Review

Campbell Biology, AP Edition, 12e

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A vesicle containing chemical molecules is seen fusing with the plasma membrane of a cell. The molecules that are released bind to proteins on the cell membrane of the adjacent cell, causing changes to these proteins. This allows for communication between cells and regulation of exchanges of cells with their environment and surrounding cells.

Summary of Key Concepts

Cell membranes control the exchange of molecules and ions between the internal and external environments of cells. Evolution has selected for a membrane structure that associates a higher energy cost in transporting hydrophilic substances than hydrophobic ones. How do you explain this, given that water is critical to life? **(BIG IDEA 2)**

Concept 7.1: Cellular membranes are fluid mosaics of lipids and proteins

- In the fluid **mosaic model**, **amphipathic**proteins are embedded in the phospholipid bilayer.
- Phospholipids and some proteins move sideways within the membrane. The unsaturated hydrocarbon tails of some phospholipids keep membranes fluid at lower temperatures, while cholesterol helps membranes resist changes in fluidity caused by temperature changes.
- Membrane proteins function in transport, enzymatic activity, signal transduction, cell-cell recognition, intercellular joining, and attachment to the cytoskeleton and extracellular matrix. Short chains of sugars linked to proteins (in **glycoproteins**) and lipids (in **glycolipids**) on the exterior side of the plasma membrane interact with surface molecules of other cells.
- Membrane proteins and lipids are synthesized in the ER and modified in the ER and Golgi apparatus. The inside and outside faces of membranes differ in molecular composition.

In what ways are membranes crucial to life?

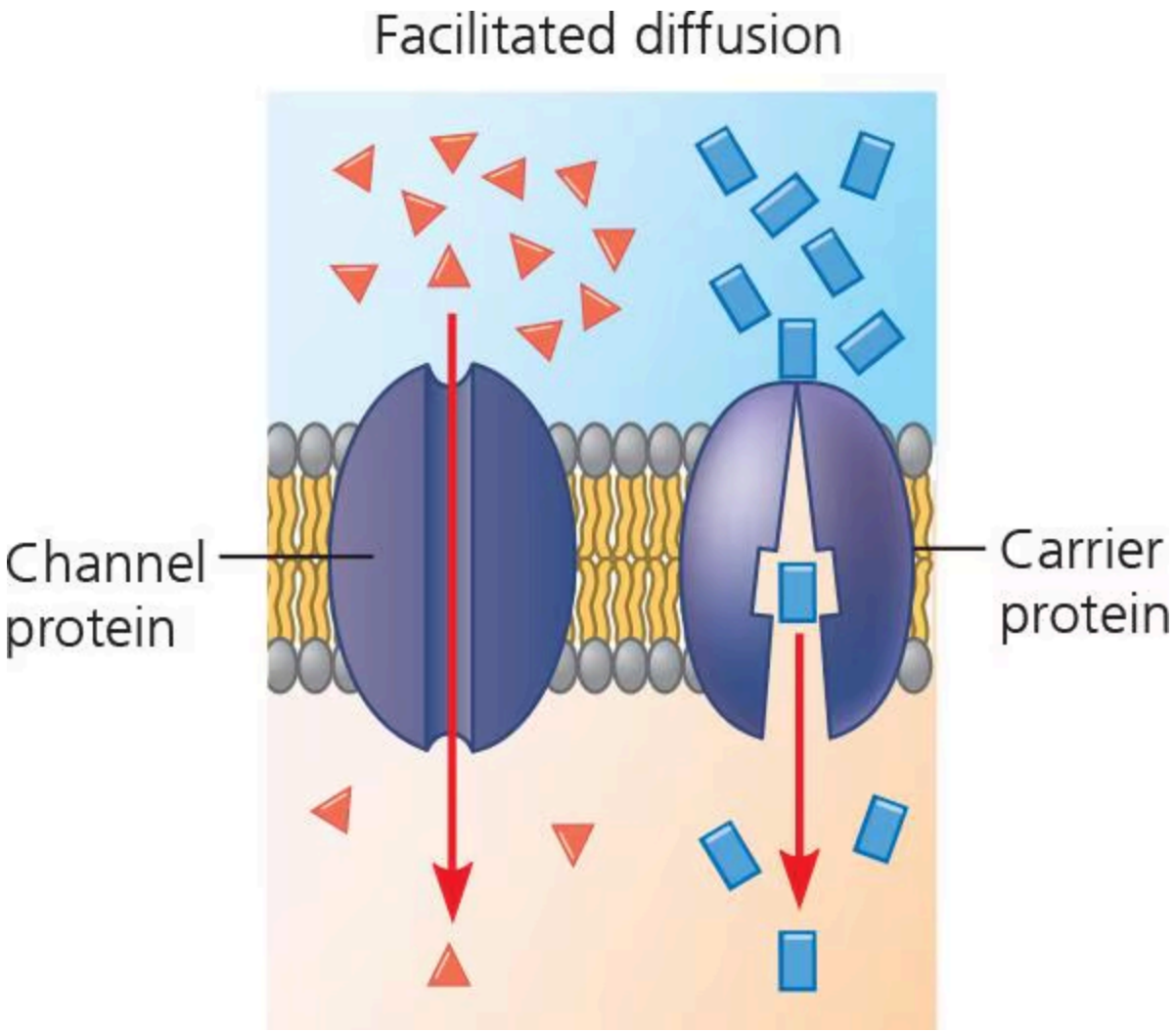
Concept 7.2: Membrane structure results in selective permeability

- A cell must exchange molecules and ions with its surroundings, a process controlled by the **selective permeability**of the plasma membrane. Hydrophobic substances are soluble in lipids and pass through membranes rapidly, whereas polar molecules and ions generally require specific **transport proteins**.

How do aquaporins affect the permeability of a membrane?

Concept 7.3: Passive transport is diffusion of a substance across a membrane with no energy investment

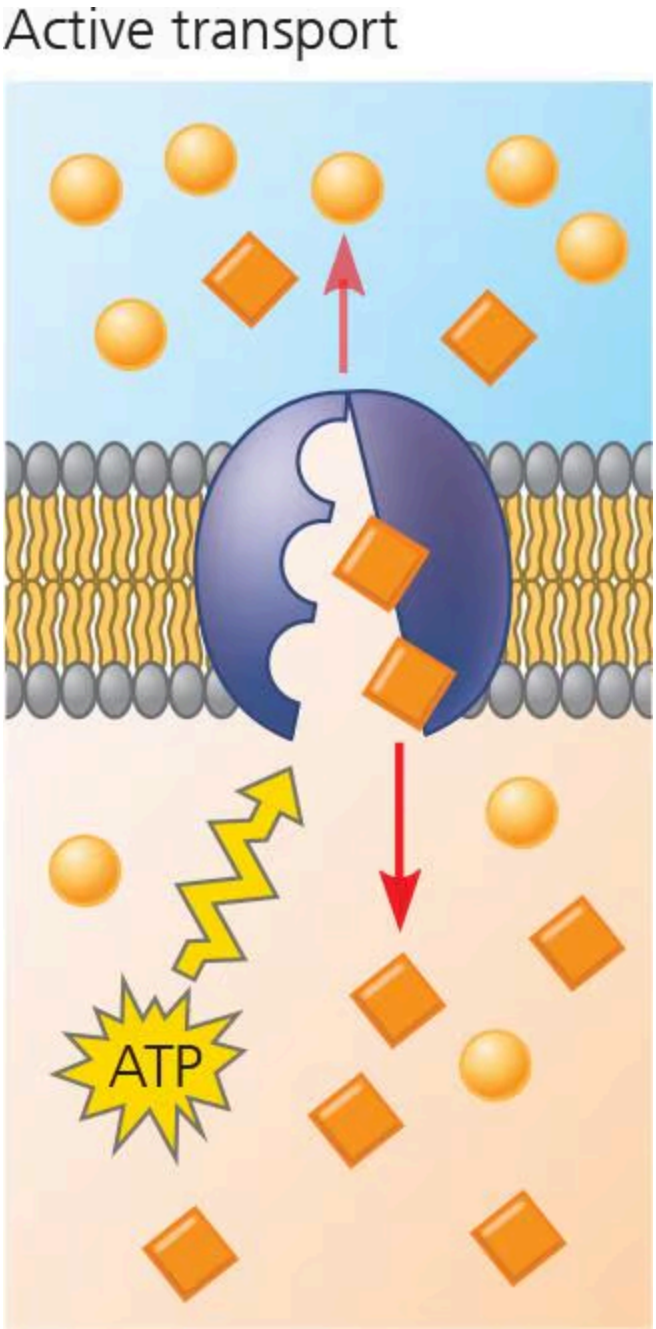
- **Diffusion** is the spontaneous movement of a substance down its **concentration gradient**. Water diffuses out of a cell (**osmosis**) if the solution outside has a higher concentration of nonpenetrating solutes (**hypertonic**); water enters if the solution has a lower solute concentration (**hypotonic**). If the concentrations are equal (**isotonic**), no net osmosis occurs. Cell survival depends on balancing water uptake and loss.
- In **facilitated diffusion**, a transport protein speeds water or solute movement down its concentration gradient across a membrane. ****Ion channels**facilitate the diffusion of ions across a membrane. Carrier proteins can undergo changes in shape that translocate bound solutes across the membrane.



What happens to a cell placed in a hypertonic solution? Describe the free water concentration inside and out.

Concept 7.4: Active transport uses energy to move solutes against their gradients

- Specific membrane proteins use energy, usually in the form of ATP, to do the work of ****__active transport__.



- Ions can have both a concentration (chemical) gradient and an electrical gradient (voltage). These gradients combine in the **electrochemical gradient**, which determines the net direction of ionic diffusion.
- **Cotransport** of two solutes occurs when a membrane protein enables the “downhill” diffusion of one solute to drive the “uphill” transport of the other.

ATP is not directly involved in the functioning of a cotransporter. Why, then, is cotransport considered active transport?

Concept 7.5: Bulk transport across the plasma membrane occurs by exocytosis and endocytosis

In ****__exocytosis__, transport vesicles migrate to the plasma membrane, fuse with it, and release their contents. In ****__endocytosis__, molecules enter cells within vesicles that pinch inward from the plasma membrane. The three types of endocytosis are ****__phagocytosis__, __pinocytosis__ and __receptor-mediated endocytosis__.

Which type of endocytosis involves the binding of specific substances in the extracellular fluid to membrane proteins? What does this type of transport enable a cell to do?